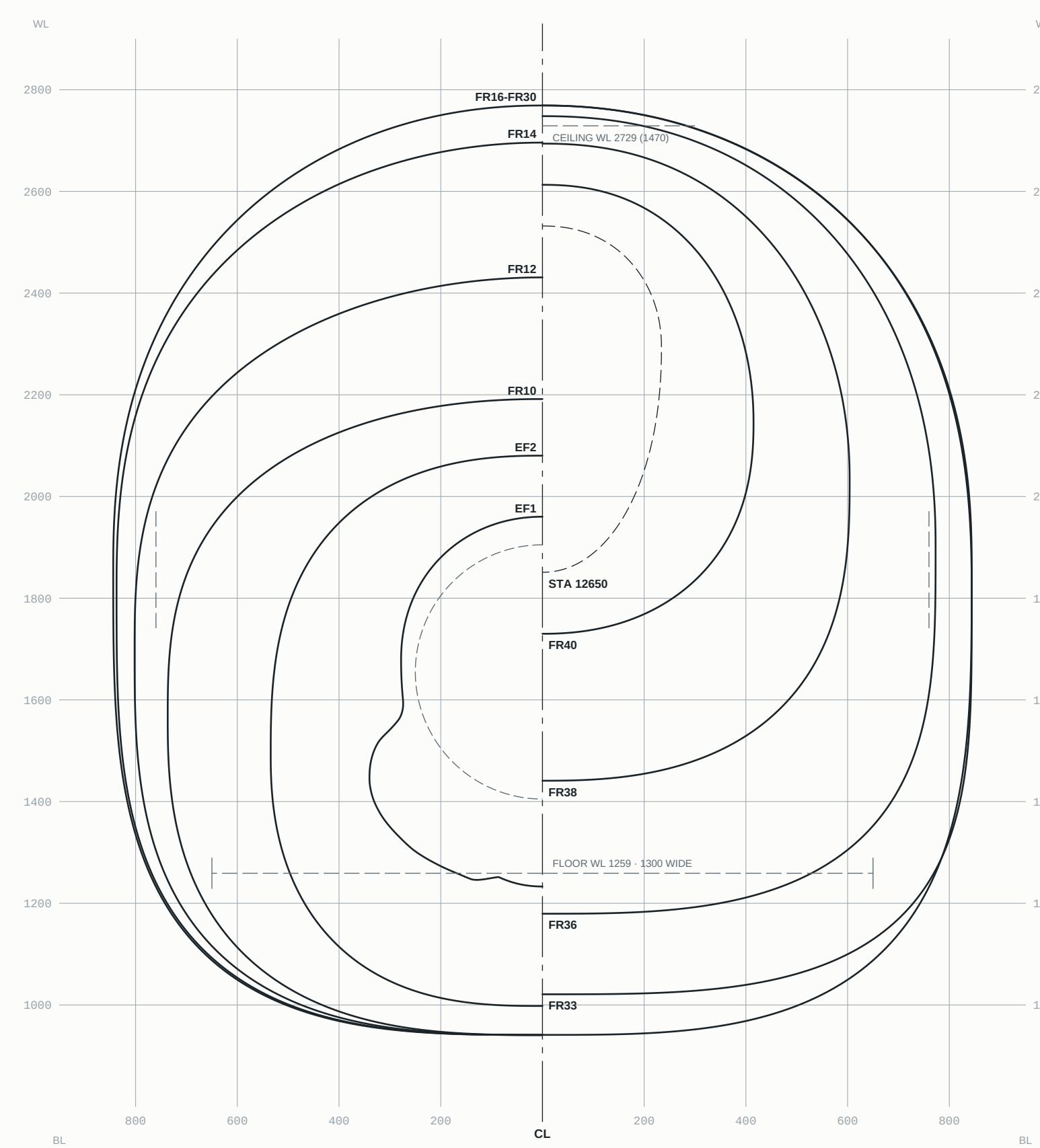
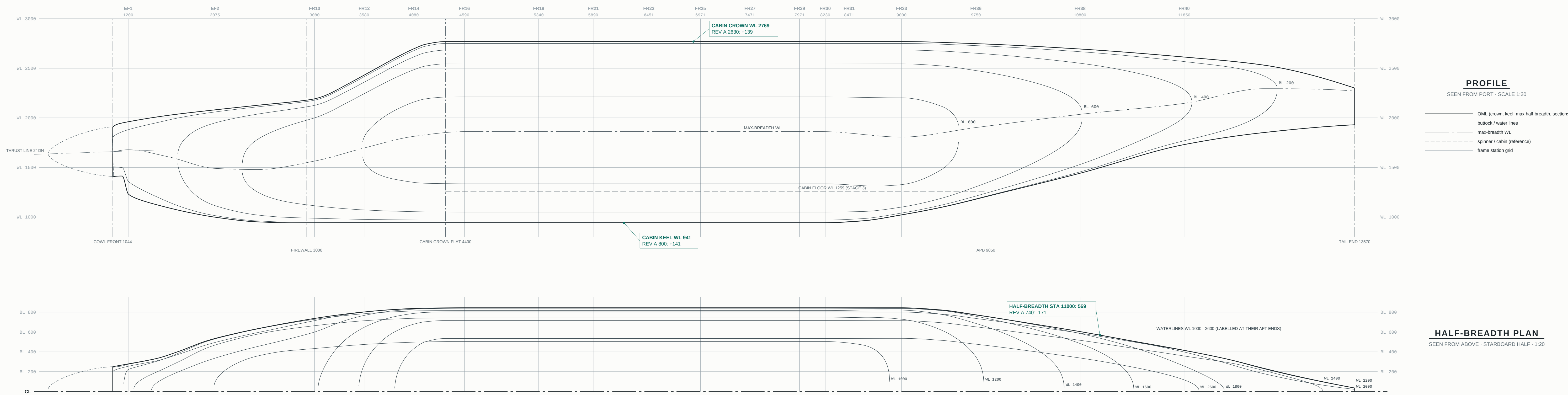




CONTROL-LINE KNOTS (model/fuselage.py)								
STA	CROWN	KEEL	HALF-BR	MAX-BR WL	n TOP	m TOP	n BOT	m BOT
1044	1905*	1405*	250*	1655*	2.00*	2.00*	2.00*	2.00*
1052	1913							
1062	1921							
1075	1929							
1090	1935							
1119	1941							
1140	1948	1413						
1180	1956							
1209		1233	278	1678	2.06	2.12	2.12	2.04
1240	1968							
1250		1201						
1329	1982							
1359		1163	305					
1420	1998							
1500		1120	336					
1540								
1600				1608	2.24	2.45	2.41	2.15
1680	2035							
1700		1848	401					
1800								
2000	2072		513	1490	2.38	2.75	2.68	2.28
2075		998						
2309		599						
2500	2126	951	632	1478	2.34	2.83	2.77	2.35
2850		941*						
3009	2178		724					
3075	2191							
3080				1564	2.16	2.81	2.83	2.53
3150	2211							
3225	2242							
3300		769						
3400	2333							
3500	2387							
3508			892	1696	2.06	2.79	2.95	2.93
3700	2407							
3900	2607	828						
4080	2696		1813	2.04	2.58	3.07	3.22	
4200	2742	841						
4400	2769*							
4600			844*	1861*	2.18*	2.35*	3.14*	3.34*
6200		941*		1861*	2.19*	2.35*	3.14*	3.34*
6600		960						
9000	2769*	1021	844*	1806	2.19	2.52	3.31	3.69
9400		1097	822					
9750	2748		773	1902	2.21	2.19	3.01	3.29
10000		1242						
10300			686					
10800	2694	1441	604	2035	2.25	2.05	2.52	2.72
11000			514					
11850	2613	1730	415	2146	2.27	2.06	2.18	2.33
12300			323					
12650	2532	1851	234	2294				
13000			151					
13570	2300*	1930*	35*	2270	2.00*	2.00*	2.00*	2.00*

mm (exponents dimensionless) · * fixed · = tied (constant cabin section)
pchip (monotone cubic) through the knots of each line

SECTION LAW (each half)

$$|y / h_w|^n + |(z - z_{mw}) / h_z|^m = 1$$

hw = half-breadth, zmw = max-breadth WL,
hz = crown - zmw (upper) or zmw - keel (lower);
n: crown / keel flatness, m: side flatness;
n = m = 2: ellipse. Section parameter t = polar
angle about (0, zmw) in normalised coordinates.

DEVIATIONS OF THE LOFT FROM THE PILATUS DRAWING (NORMAL DISTANCE, mm)						
DATASET	N	RMS	MAX	REV A MAX	REV A MAX	WORST AT
PROFILE CROWN	794	1.0	9.4	109.5	141.1	STA 1060
PROFILE KEEL	749	2.1	17.2	179.2	339.7	STA 1060
HALF-BREADTH	1164	0.9	3.1	90.0	230.7	STA 1240
FIN-ROOT LINE	161	1.6	3.4	118.1	137.5	STA 12750
LOWER SILHOUETTE	286	-	0.0	-	343.3	
EP1 1200	68	4.0	14.5	68.0	95.8	BL 254 WL 1559
EP2 2075	133	0.7	1.5	72.2	134.7	BL 292 WL 1042
EP3 3200	133	1.8	3.5	26.7	54.2	BL 475 WL 1060
EP12 3580	133	2.8	6.0	74.7	93.1	BL 731 WL 2096
FR14 4080	144	3.2	6.9	56.7	130.5	BL 308 WL 961
FR16-30 4550-8230	173	2.3	8.2	77.8	139.1	BL 337 WL 965
FR33 9000	223	4.3	11.4	90.2	212.1	BL 818 WL 1444
FR38 9750	189	1.6	3.4	116.6	295.9	BL 745 WL 1550
FR38 10000	69	0.9	2.5	20.3	344.3	BL 0 WL 2096
FR40 11850	116	0.5	1.2	207.4	364.0	BL 403 WL 2263
FR40 11851 (SHEET 2)	840	4.3	12.1	66.0	139.8	FR31 BL 0 WL 939

Targets: profile / half-breadth RMS ≤ 8 , max ≤ 20 ; every frame section max ≤ 20 - ALL MET. REV A = the model before this refit

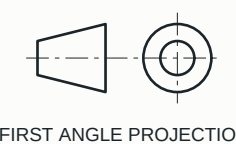
CABIN SECTION NUMBERS		
CABIN (STAGE 3)		mm
crown / keel WL (STA 4600-8200)		2769 / 941
OML depth / max breadth		1828 / 1838
max-breadth WL		1861
ceiling WL (crown - 40)		2729
floor WL (ceiling - 1470)		1259
floor WL change vs rev A (1120)		+139
OML width at the floor		1545
floor width avail. (- 2 x 85)		1375 => 1390
max inner width (- 2 x 85)		1518 - 1520

NOTES

1. Drawn from the left side of models/EU560stage by (knot tabs + section law), not from the mesh. Stations = STA m aft of the datum (3000 ahead of the firewall), WL above the static ground line; BL from the centre line, + starboard.
2. Knot values filled by least squares (Law560s_fit) to the registered Platons NGX model drawing 190,40,40,42 side crown / visible knee, two side crown / visible knee, frame sections EF1-F40 (sheet 1) and sheet 2 z-platons chain section, with smoothness regularisation. Deviations = normal distribution.
3. Fixed: spinner lip ST4300, prop axis WL (drawings at the disc) (prop clearance 320 kept; the drawing shows 136; thus side crown + 2 deg right-hand), so the spinner lip is at WL 1636 and the spinner base centre 4 mm above the cow-rl front (R 250 at STA 1044, Stage 3; firewall STA 3000, tail cone closure STA 13570 at the lower end of the rudder leading edge; 14,600 overall length is set by the tail bulge).
4. Not part of this CMM (Stage 3 parts): chin-ribs encased dorsal fin, ventral strakes, ventral leading under the tail cone (FWA 1900, lower silhouette aft of STA 11900), wing-root fairing. Aft of the rudder leading edge (STA 12500 at the keel) the tail cone lies inside the rudder / or trim is trimmed there.
5. Changes vs rev A: cabin crown ST230 > 2769, leg root 800 > 941 (flat from STA 2800), aft body slimmer (half-frame STA 11000 740 > 569), lat end 14360 > 13570, spinner R 230 > 250, cow-rl front 950 > 1044.

PILATUS PC-12 PRO

FUSELAGE LINES PLAN



DWG NO	REV	SCALE	SHEET	UNITS
PC12-L1	A	AS SHOWN	A1 · 1 OF 1	mm
DRAWN Claude - loftkit parametric model				DATE 2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION